WORKSHEET 4.5

Related Rates: Solving

Unit 4: Derivatives in Context

Key ideas

- A length that does not change (a ladder, a rope of fixed length) is a **constant**: put its value in before differentiating.
- A length that changes must stay a **variable** until after differentiating.
- For a cone, use similar triangles to write r in terms of h (or the reverse) before differentiating.
- For an angle, choose the trigonometric ratio that involves only one changing length if you can.

Worked examples

Example 1 · A ladder

A 17 m ladder leans against a wall. Its foot is pulled away from the wall at 1.5 m/s. How fast is the top moving when the foot is 8 m from the wall?

Solution. $x^2 + y^2 = 289$, so $x\dfrac{dx}{dt} + y\dfrac{dy}{dt} = 0$. When $x = 8$, $y = 15$: $8(1.5) + 15\dfrac{dy}{dt} = 0$.

The top is sliding down at 0.8 m/s.

Example 2 · A cone

Water is poured at 2 m^3/min into an inverted cone of height 6 m and top radius 4 m. How fast is the water level rising when the water is 3 m deep?

Solution. $r = \dfrac{2h}{3}$, so $V = \dfrac{1}{3}\pi\left(\dfrac{2h}{3}\right)^2 h = \dfrac{4\pi h^3}{27}$ and $\dfrac{dV}{dt} = \dfrac{4\pi h^2}{9}\dfrac{dh}{dt}$. At $h = 3$: $2 = 4\pi\dfrac{dh}{dt}$.

$\dfrac{1}{2\pi}$ **m/min.**

Example 3 · A shadow

A person 1.8 m tall walks at 1.2 m/s away from a lamp post 5.4 m high. How fast is the length of the shadow increasing?

Solution. With x the distance from the post and s the shadow: $\dfrac{x+s}{5.4} = \dfrac{s}{1.8}$, so $s = \dfrac{x}{2}$.

0.6 **m/s.**

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Practice

CORE

1. A circular ripple has radius increasing at 2 cm/s. How fast is its area increasing when the radius is 10 cm?

2. The edge of a cube is increasing at 0.1 cm/s. How fast is its volume increasing when the edge is 5 cm?

3. A 15 m ladder leans against a wall. Its foot moves away from the wall at 2 m/s. How fast is the top moving when the foot is 9 m from the wall?

4. A spherical balloon is inflated at 64π cm^3/s. How fast is its radius increasing when $r = 2$ cm?

EXAM LEVEL

5. Sand is poured at 10 m^3/min onto a conical pile whose height always equals its radius. How fast is the height increasing when the pile is 5 m high?

6. Two cars leave the same point at the same time. One travels east at 60 km/h and the other north at 80 km/h. How fast is the distance between them increasing after 2 hours?

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7. A camera on the ground is 300 m from a launch pad. A rocket rises vertically at 150 m/s. How fast is the camera's angle of elevation changing when the rocket is 400 m high?

8. Water is pumped at 18π m^3/h into a cylindrical tank of radius 3 m. How fast is the water level rising?

Question 9 · Free-response practice

A boat is pulled toward a dock by a rope that passes through a ring on the dock 5 m higher than the bow of the boat. The rope is pulled in at 2 m/s. Let x be the horizontal distance from the boat to the dock and z the length of rope between the boat and the ring.

(a) Write an equation relating x and z, and differentiate it with respect to time.

(b) How fast is the boat approaching the dock when 13 m of rope are out?

(c) Let θ be the angle between the rope and the water. How fast is θ changing at that moment?

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Worked solutions

1. $\dfrac{dA}{dt} = 2\pi r\dfrac{dr}{dt} = 40\pi$ cm^2/s.

2. $3(25)(0.1) = 7.5$ cm^3/s.

3. $y = 12$. $9(2) + 12\dfrac{dy}{dt} = 0$, so the top slides down at 1.5 m/s.

4. $64\pi = 4\pi(4)\dfrac{dr}{dt}$, so 4 cm/s.

5. $r = h$, so $V = \dfrac{\pi h^3}{3}$ and $\dfrac{dV}{dt} = \pi h^2\dfrac{dh}{dt}$: $10 = 25\pi\dfrac{dh}{dt}$, giving $\dfrac{2}{5\pi}$ m/min.

6. $x = 120$, $y = 160$, $z = 200$. $200\dfrac{dz}{dt} = 120(60) + 160(80)$, so 100 km/h.

7. $\tan\theta = \dfrac{y}{300}$, so $\sec^2\theta\,\dfrac{d\theta}{dt} = \dfrac{150}{300}$. With $\sec^2\theta = 1 + \dfrac{16}{9} = \dfrac{25}{9}$: $\dfrac{d\theta}{dt} = 0.18$ rad/s.

8. $V = 9\pi h$, so $18\pi = 9\pi\dfrac{dh}{dt}$: 2 m/h.

9. (a) $z^2 = x^2 + 25$, so $z\dfrac{dz}{dt} = x\dfrac{dx}{dt}$.

   (b) $x = 12$ and $\dfrac{dz}{dt} = -2$: $13(-2) = 12\dfrac{dx}{dt}$, so the boat approaches at $\dfrac{13}{6}$ m/s.

   (c) $\sin\theta = \dfrac{5}{z}$, so $\cos\theta\,\dfrac{d\theta}{dt} = -\dfrac{5}{z^2}\dfrac{dz}{dt}$. With $\cos\theta = \dfrac{12}{13}$: $\dfrac{12}{13}\dfrac{d\theta}{dt} = \dfrac{10}{169}$, so $\dfrac{d\theta}{dt} = \dfrac{5}{78}$ rad/s (increasing).

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